Annotated BHTTP1 Request

Live TCP capture | 96 bytes | Request ID 1

These bytes were emitted by bcurl and received by bserve. Hexadecimal offsets start at the beginning of each frame. Every byte is accounted for in the field map.

```
0000  00 00 00 58 01 00 00 01 01 00 0a 2f 68 65 6c 6c  |...X......./hell|
0010  6f 2e 74 78 74 06 01 00 0f 31 32 37 2e 30 2e 30  |o.txt....127.0.0|
0020  2e 31 3a 35 30 37 39 30 02 00 07 62 63 75 72 6c  |.1:50790...bcurl|
0030  2f 31 03 00 03 2a 2f 2a 07 00 0a 6b 65 65 70 2d  |/1...*/*...keep-|
0040  61 6c 69 76 65 09 00 01 31 0a 00 14 6e 65 74 77  |alive...1...netw|
0050  6f 72 6b 2d 61 72 63 68 69 74 65 63 74 75 72 65  |ork-architecture|
```

Hex offsets	Field	Decoded meaning
0000-0003	Payload length	88 bytes; header excluded
0004	Type	1 REQUEST
0005	Flags	0; reserved bits clear
0006-0007	Request ID	1
0008	Method	1 GET
0009-000a	Path length	10
000b-0014	UTF-8 path	/hello.txt
0015	Header count	6
0016-0027	Header 1	ID 1 = host; valueLen 15; value 127.0.0.1:50790
0028-0031	Header 2	ID 2 = user-agent; valueLen 7; value bcurl/1
0032-0037	Header 3	ID 3 = accept; valueLen 3; value */*
0038-0044	Header 4	ID 7 = connection; valueLen 10; value keep-alive
0045-0048	Header 5	ID 9 = x-request-id; valueLen 1; value 1
0049-005f	Header 6	ID 10 = x-course; valueLen 20; value network-architecture

Annotated BHTTP1 Response

Live TCP capture | 139 bytes | Request ID 1

Status 200 returns the exact hello.txt bytes. The literal header x-file-name demonstrates the fallback representation; all ten predefined IDs appear across the request and response.

```
0000  00 00 00 83 02 00 00 01 00 c8 08 04 00 0a 74 65  |..............te|
0010  78 74 2f 70 6c 61 69 6e 05 00 02 32 30 06 00 08  |xt/plain...20...|
0020  62 73 65 72 76 65 2f 31 07 00 0a 6b 65 65 70 2d  |bserve/1...keep-|
0030  61 6c 69 76 65 08 00 08 6e 6f 2d 73 74 6f 72 65  |alive...no-store|
0040  09 00 01 31 0a 00 14 6e 65 74 77 6f 72 6b 2d 61  |...1...network-a|
0050  72 63 68 69 74 65 63 74 75 72 65 00 0b 78 2d 66  |rchitecture..x-f|
0060  69 6c 65 2d 6e 61 6d 65 00 09 68 65 6c 6c 6f 2e  |ile-name..hello.|
0070  74 78 74 00 00 00 14 48 65 6c 6c 6f 2c 20 62 69  |txt....Hello, bi|
0080  6e 61 72 79 20 48 54 54 50 21 0a                 |nary HTTP!.|
```

Hex offsets	Field	Decoded meaning
0000-0003	Payload length	131 bytes; header excluded
0004	Type	2 RESPONSE
0005	Flags	0; reserved bits clear
0006-0007	Request ID	1
0008-0009	Status	200
000a	Header count	8
000b-0017	Header 1	ID 4 = content-type; valueLen 10; value text/plain
0018-001c	Header 2	ID 5 = content-length; valueLen 2; value 20
001d-0027	Header 3	ID 6 = server; valueLen 8; value bserve/1
0028-0034	Header 4	ID 7 = connection; valueLen 10; value keep-alive
0035-003f	Header 5	ID 8 = cache-control; valueLen 8; value no-store
0040-0043	Header 6	ID 9 = x-request-id; valueLen 1; value 1
0044-005a	Header 7	ID 10 = x-course; valueLen 20; value network-architecture
005b-0072	Header 8	ID 0; nameLen 11; name x-file-name; valueLen 9; value hello.txt
0073-0076	Body length	20
0077-008a	Body bytes	Hello, binary HTTP!\n

Byte verification The output and served file have identical SHA-256 hashes:
e39a22ac0e29905c83bc5e51f209b9249aa4db8db722e99c8605dd1624eda3f2